

22/2/21

venerdì 10 giugno 2022 16:34

dimostrare che il seguente linguaggio non è CFL:

$$L = \{ 0^m 1 0^m 1 0^q \mid m, m > 0 \text{ e } q = \max(m, m) \}$$

Svilg

$$S = 0^p 1 0^p 1 0^p$$

$$1) S = \underbrace{00000 \dots 0}_p 1 \underbrace{000000 \dots 0}_p 1 \underbrace{000000 \dots 0}_p$$

vxy

$$\left[\begin{array}{l} v=0 \\ x=0 \\ y=0 \end{array} \quad \begin{array}{l} v=0^+ \\ x=0^+ \\ y=0^+ \end{array} \right] \text{UP: } 0^S 1 0^p 1 0^p, \quad S \neq p, S > p, \notin L$$

$$2) S = \underbrace{00000 \dots 0}_p 1 \underbrace{000000 \dots 0}_p 1 \underbrace{000000 \dots 0}_p$$

vxy

$$\left[\begin{array}{l} v=0 \\ x=0 \\ y=0 \end{array} \quad \begin{array}{l} v=0^+ \\ x=0^+ \\ y=0^+ \end{array} \right] \text{UP: } 0^p 1 0^S 1 0^p, \quad S > p, p \neq \max(S, p), \notin L!$$

$$3) S = \underbrace{00000 \dots 0}_p 1 \underbrace{000000 \dots 0}_p 1 \underbrace{000000 \dots 0}_p$$

vxy

$$\left[\begin{array}{l} v=0 \\ x=0 \\ y=0 \end{array} \quad \begin{array}{l} v=0^+ \\ x=0^+ \\ y=0^+ \end{array} \right] \text{UP: } 0^p 1 0^p 1 0^S, \quad S > p, S \neq p, \notin L$$

$$4) S = \underbrace{00000 \dots 0}_p \underbrace{1}_{vxy} \underbrace{000000 \dots 0}_p 1 \underbrace{00000 \dots 0}_p$$

$$\left[\begin{array}{l} v=0 \\ x=1 \\ y=0 \end{array} \right] \quad \left[\begin{array}{l} v=0^+ \\ x=1 \\ y=0^+ \end{array} \right] \quad \text{up: } 0^S 1 0^S 1 0^P, \quad p \neq \max(S, S) \notin L$$

$$5) S = \underbrace{00000 \dots 0}_p 1 \underbrace{000000 \dots 0}_p \underbrace{1}_{vxy} \underbrace{00000 \dots 0}_p$$

$$\left[\begin{array}{l} v=0 \\ x=0 \\ y=0 \end{array} \right] \quad \left[\begin{array}{l} v=0^+ \\ x=0^+ \\ y=0^+ \end{array} \right] \quad \text{up: } 0^P 1 0^S 1 0^S \in$$

$$\text{down: } 0^P 1 0^S 1 0^S, \quad S < P, \quad S \neq \max(S, P) \notin L!$$

IS NOT CFL.